

Tensorial Multidimensional Pandrosion

Averaged Jacobian Divided Differences, Bilinear Residual Laws, and Tensor-Flat Charts

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Abstract

The scalar anchored Pandrosion map replaces the monomial denominator by an anchored divided difference and makes a root into a fixed point whose multiplier is read from a secant slope. This paper develops the corresponding local theory for analytic systems $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$. The scalar divided difference is replaced by the averaged Jacobian

$$\mathcal{Q}_F(a, s) = \int_0^1 DF(s + t(a - s)) dt, \quad F(a) - F(s) = \mathcal{Q}_F(a, s)(a - s),$$

and the multidimensional anchored map is

$$P_{F,a}(s) = a - \mathcal{Q}_F(a, s)^{-1}F(a).$$

At a simple zero r , with $A = DF(r)$ and $B = D^2F(r)$, the local two-point law is

$$P_{F,a}(s) - r = \frac{1}{2}A^{-1}B[a - r, s - r] + \mathcal{O}((\|a - r\| + \|s - r\|)^3).$$

Thus the raw fixed-anchor map has a linear multiplier controlled by the anchor vector through the bilinear tensor $A^{-1}B$. Under a chart ϕ , the tensor $D^2(G \circ \phi)(0)$ is the multidimensional analogue of scalar curvature. A root-normalized chart is first-order tensor-flat precisely when

$$D^2(G \circ \phi)(0) = 0.$$

The quadratic jet realizing this cancellation is the second-order local inverse jet of G . More generally, an m -flat chart satisfying $G(\phi(u)) = u + \mathcal{O}(\|u\|^{m+1})$ makes the local multiplier $\mathcal{O}(\|a\|^m)$. Numerical figures for a two-dimensional analytic system illustrate the multiplier ladder, the anisotropy of anchor directions, and the enlarged low-error region produced by tensor cancellation.

One-sentence summary. Multidimensional Pandrosion replaces scalar slopes by averaged Jacobians; its first residual tensor is $\frac{1}{2}A^{-1}D^2F(r)$, and tensor-flat charts cancel that tensor before iteration begins.

1 Introduction

The scalar theory of anchored divided differences shows that a fixed anchor turns a simple zero into a fixed point and that the residual multiplier is the mismatch between a tangent slope and an anchor-root secant slope. The general analytic paper proves the scalar local product law

$$P_{F,a}(s) - r \sim \frac{F''(r)}{2F'(r)}(a - r)(s - r),$$

while the optimal-chart and higher-order papers show that one can reduce this residual factor by flattening the transformed equation in a suitable coordinate.

The purpose of the present paper is to move the same idea to systems. The correct object is not a scalar denominator but a matrix-valued divided difference. For a system $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$, the natural average along the segment from s to a is

$$\mathcal{Q}_F(a, s) = \int_0^1 DF(s + t(a - s)) dt.$$

This operator satisfies the exact identity

$$F(a) - F(s) = \mathcal{Q}_F(a, s)(a - s),$$

which is the multidimensional replacement for the scalar divided difference.

The key observation is that the first residual obstruction is tensorial. If r is a simple zero, $A = DF(r)$, and $B = D^2F(r)$, then

$$P_{F,a}(s) - r = \frac{1}{2}A^{-1}B[a - r, s - r] + \text{higher order}.$$

This formula is the multivariate version of the scalar fixed-anchor secant product law. It is also the formula behind the practical finite-slope engines: good anchors and well-conditioned averaged Jacobians are the natural geometric data.

Main contributions. The paper proves the following local facts.

- (i) The averaged-Jacobian map has exactly the same fixed-point property as the scalar anchored map.
- (ii) The leading error is the bilinear tensor $\frac{1}{2}A^{-1}D^2F(r)[a - r, s - r]$.
- (iii) In a chart ϕ , first-order tensor cancellation is the coordinate condition

$$D^2(G \circ \phi)(0) = 0.$$

- (iv) The unique quadratic jet satisfying this condition is the second-order inverse jet

$$\phi_2(u) = \rho + A^{-1}u - \frac{1}{2}A^{-1}B[A^{-1}u, A^{-1}u].$$

- (v) Higher-order inverse jets yield a tensorial m -flat hierarchy

$$G(\phi_m(u)) = u + \mathcal{O}(\|u\|^{m+1}) \implies \|DP_{\phi_m, a}(0)\| = \mathcal{O}(\|a\|^m).$$

2 Averaged Jacobian divided differences

Let $U \subset \mathbb{C}^n$ be a convex neighborhood and let $F : U \rightarrow \mathbb{C}^n$ be analytic.

Definition 2.1 (Averaged Jacobian divided difference). For $a, s \in U$, define

$$\mathcal{Q}_F(a, s) = \int_0^1 DF(s + t(a - s)) dt. \tag{1}$$

By the fundamental theorem of calculus,

$$F(a) - F(s) = \mathcal{Q}_F(a, s)(a - s). \tag{2}$$

Thus $\mathcal{Q}_F(a, s)$ is a canonical matrix-valued divided difference along the segment from s to a .

Definition 2.2 (Multidimensional anchored Pandrosion). Assume $F(a) \neq 0$ and $\mathcal{Q}_F(a, s)$ is invertible. The anchored multidimensional Pandrosion map is

$$P_{F,a}(s) = a - \mathcal{Q}_F(a, s)^{-1}F(a). \quad (3)$$

Equivalently, using (2),

$$P_{F,a}(s) = s - \mathcal{Q}_F(a, s)^{-1}F(s). \quad (4)$$

Proposition 2.3 (Fixed points). *Let r be a zero of F , $F(a) \neq 0$, and $\mathcal{Q}_F(a, r)$ be invertible. Then*

$$P_{F,a}(r) = r.$$

Conversely, if $P_{F,a}(s) = s$ and $\mathcal{Q}_F(a, s)$ is invertible, then $F(s) = 0$.

Proof. If $F(r) = 0$, then (2) gives $F(a) = \mathcal{Q}_F(a, r)(a - r)$, hence

$$P_{F,a}(r) = a - \mathcal{Q}_F(a, r)^{-1}\mathcal{Q}_F(a, r)(a - r) = r.$$

Conversely, $P_{F,a}(s) = s$ gives $\mathcal{Q}_F(a, s)(a - s) = F(a)$. By (2), the left side is $F(a) - F(s)$, so $F(s) = 0$. $\square$

3 The tensor product law

Let r be a simple zero, and translate coordinates so $r = 0$. Write

$$F(w) = Aw + \frac{1}{2}B[w, w] + \frac{1}{6}C[w, w, w] + \mathcal{O}(\|w\|^4),$$

where $A = DF(0)$ is invertible and $B = D^2F(0)$ is a symmetric vector-valued bilinear map.

Theorem 3.1 (Bilinear local residual law). *Let $a = \alpha$ and $s = \sigma$ be close to 0. Then*

$$P_{F,a}(s) = \frac{1}{2}A^{-1}B[\alpha, \sigma] + \mathcal{O}(\|\alpha\|^2\|\sigma\| + \|\alpha\|\|\sigma\|^2 + \|\sigma\|^3). \quad (5)$$

Consequently the derivative of the fixed-anchor map at the root is

$$DP_{F,a}(0)v = \frac{1}{2}A^{-1}B[a, v] + \mathcal{O}(\|a\|^2\|v\|). \quad (6)$$

Proof. From the integral definition,

$$\mathcal{Q}_F(\alpha, \sigma)v = Av + \frac{1}{2}B[\alpha + \sigma, v] + \mathcal{O}((\|\alpha\|^2 + \|\alpha\|\|\sigma\| + \|\sigma\|^2)\|v\|).$$

Therefore

$$\mathcal{Q}_F(\alpha, \sigma)^{-1} = A^{-1} - \frac{1}{2}A^{-1}B[\alpha + \sigma, A^{-1}(\cdot)] + \mathcal{O}((\|\alpha\| + \|\sigma\|)^2).$$

Using the secant form,

$$P_{F,a}(s) = \sigma - \mathcal{Q}_F(\alpha, \sigma)^{-1}F(\sigma),$$

and

$$F(\sigma) = A\sigma + \frac{1}{2}B[\sigma, \sigma] + \mathcal{O}(\|\sigma\|^3).$$

Substitution yields

$$\mathcal{Q}_F(\alpha, \sigma)^{-1}F(\sigma) = \sigma + \frac{1}{2}A^{-1}B[\sigma, \sigma] - \frac{1}{2}A^{-1}B[\alpha + \sigma, \sigma] + \text{higher order} = \sigma - \frac{1}{2}A^{-1}B[\alpha, \sigma] + \text{higher order}.$$

Subtracting from σ gives (5). Differentiating with respect to σ gives (6). $\square$

Remark 3.2. The scalar formula is recovered by taking $A = F'(0)$ and $B = F''(0)$. Then (5) becomes

$$P_{F,a}(s) = \frac{F''(0)}{2F'(0)}as + \dots$$

4 Tensor-flat charts

Let $G : \Omega \rightarrow \mathbb{C}^n$ be analytic and let ρ be a simple zero. Let ϕ be a local biholomorphic chart with $\phi(0) = \rho$, and set

$$F_\phi(u) = G(\phi(u)).$$

Definition 4.1 (Root-normalized and tensor-flat chart). A chart ϕ is root-normalized if

$$D(G \circ \phi)(0) = I.$$

It is first-order tensor-flat if, in addition,

$$D^2(G \circ \phi)(0) = 0.$$

More generally, it is m -flat if

$$G(\phi(u)) = u + \mathcal{O}(\|u\|^{m+1}).$$

Theorem 4.2 (Quadratic tensor cancellation). *Let $A = DG(\rho)$ and $B = D^2G(\rho)$. The quadratic chart jet*

$$\phi_2(u) = \rho + A^{-1}u - \frac{1}{2}A^{-1}B[A^{-1}u, A^{-1}u] \quad (7)$$

is root-normalized and first-order tensor-flat:

$$G(\phi_2(u)) = u + \mathcal{O}(\|u\|^3).$$

It is the unique quadratic jet with these properties.

Proof. Write $\phi(u) = \rho + Lu + \frac{1}{2}H[u, u] + \mathcal{O}(\|u\|^3)$. Root normalization forces $AL = I$, so $L = A^{-1}$. The quadratic term of $G \circ \phi$ is

$$\frac{1}{2}AH[u, u] + \frac{1}{2}B[Lu, Lu].$$

Setting it equal to zero gives

$$H[u, u] = -A^{-1}B[A^{-1}u, A^{-1}u],$$

which is exactly (7). Uniqueness follows because A is invertible. □

Corollary 4.3 (Tensor-flat multiplier order). *If ϕ is first-order tensor-flat, then*

$$\|DP_{\phi,a}(0)\| = \mathcal{O}(\|a\|^2).$$

If ϕ is m -flat, then

$$\|DP_{\phi,a}(0)\| = \mathcal{O}(\|a\|^m).$$

Proof. Apply [Theorem 3.1](#) to $F_\phi = G \circ \phi$. If $D^2F_\phi(0) = 0$, the bilinear tensor vanishes and the next term is quadratic in the anchor. The m -flat statement follows by the same Taylor expansion degree by degree. □

5 Figures: a two-dimensional test system

Consider

$$G_1(x, y) = x + 0.55x^2 + 0.25xy - 0.15y^2,$$

$$G_2(x, y) = y - 0.20x^2 + 0.35xy + 0.50y^2.$$

The root is 0 and $DG(0) = I$. The tensor-flat quadratic chart is therefore

$$\phi_2(u) = u - Q_2(u),$$

where Q_2 is the quadratic part of G .

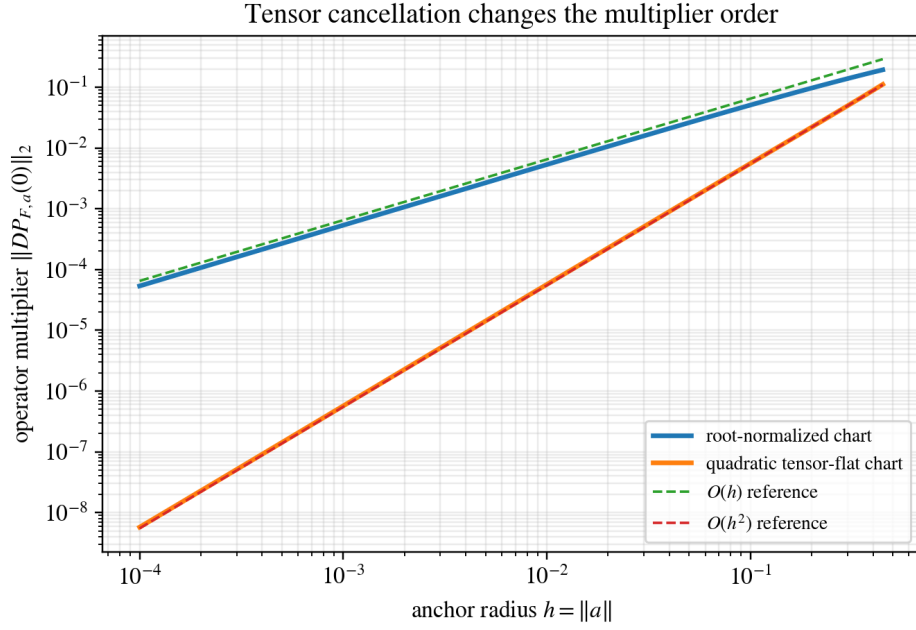


Figure 1: Multiplier norm versus anchor radius. The root-normalized chart has slope 1 on a log-log scale, while the tensor-flat chart has slope 2, confirming the tensor cancellation law.

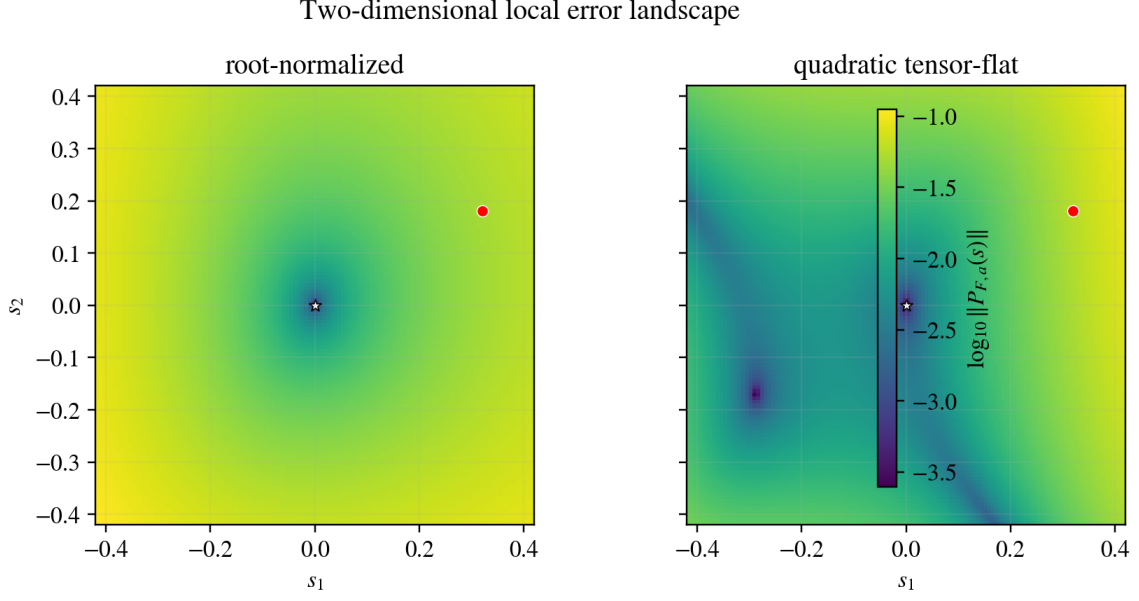


Figure 2: Local error landscape for a fixed anchor. Darker regions correspond to smaller $\|P_{F,a}(s)\|$. Tensor cancellation enlarges the low-error region around the root.

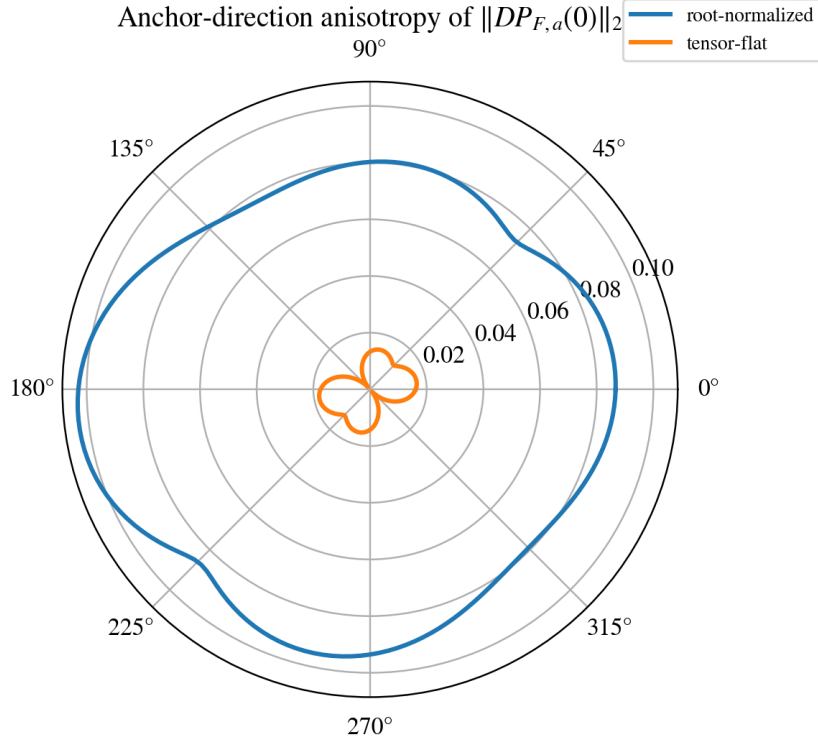


Figure 3: The bilinear tensor $A^{-1}B[a, \cdot]/2$ makes the multiplier anisotropic in the anchor direction. Tensor-flat coordinates suppress the leading anisotropic lobe.

6 Discussion

The multidimensional theory clarifies what the finite-slope engines are approximating. They are not merely using random secants; they are sampling averaged Jacobians whose local defect is

governed by a symmetric bilinear tensor. The practical lessons are direct:

- (1) choose anchors that reduce $\|A^{-1}B[a, \cdot]\|$;
- (2) reject probes for which $\mathcal{Q}_F(a, s)$ is poorly conditioned;
- (3) use chart transformations to cancel the quadratic tensor when derivative information is available or estimable.

This is the natural bridge between pure Pandrosion engines and higher-order geometric preconditioning. The scalar curvature condition becomes a tensor cancellation condition. Higher-order inverse jets become symmetric multilinear normal forms.

7 Conclusion

Multidimensional Pandrosion is governed by an averaged Jacobian divided difference. At a simple zero, the first nontrivial residual is the bilinear tensor

$$\frac{1}{2}(DF(r))^{-1}D^2F(r)[a - r, s - r].$$

A chart can cancel this tensor, and the quadratic inverse jet gives the unique local cancellation. This provides a clean tensorial continuation of the scalar anchored divided-difference program.

References

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